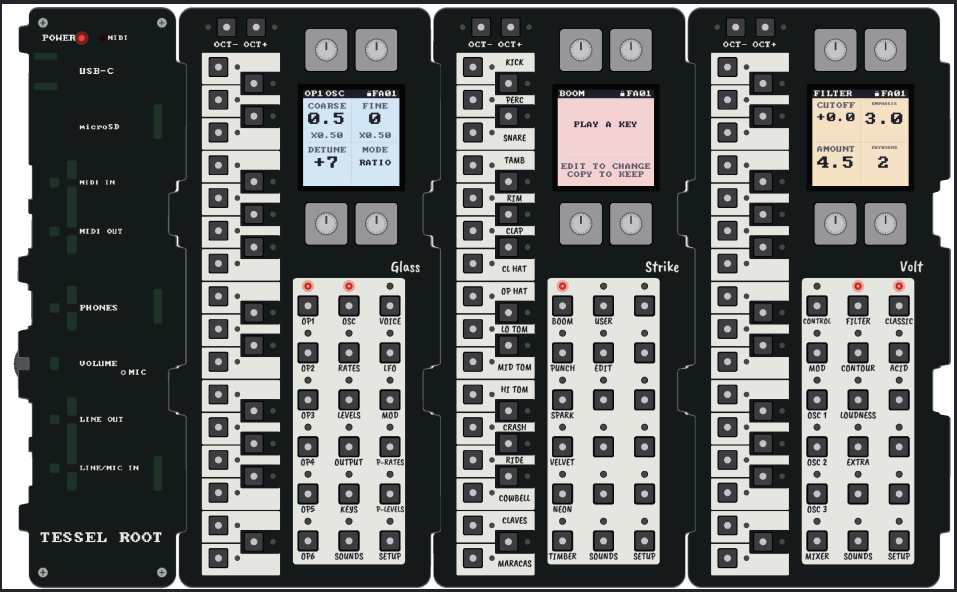
# TESSEL

## Quick Start




UTOPIAN INSTRUMENTS

# Hello

A Tessel is a row of small instruments that snap together side by side. The **Root** sits on the left: it takes the power and gives out the sound. Each unit to its right can become any one of ten instruments, from an FM synth to a drum machine or a sequencer. You choose, and you can change your mind whenever you like.

This booklet gets you from the box to your first sounds. The full manuals are free to download (page 7).

## In the box

The circuit board, the top panel, the base, and the clear side walls (two on a unit, one on the Root); the screen and a strip of double-sided tape (units only); five screws, five posts, four domed nuts and one plain nut; a screwdriver and a spanner.

## You'll also need

- **A USB-C power supply or power bank, and a cable.** A supply rated 3 A runs a full row of eight units; an older charger with a USB-A end may run only two or three.
- **Headphones**, or speakers with a 3.5 mm lead.
- **A fine permanent marker**, to write each unit's button names on its white panel.
- Later, perhaps, **a microSD card**. Taste and Grit keep about 40 seconds of your own samples on the Root; for more, the Root needs a card.

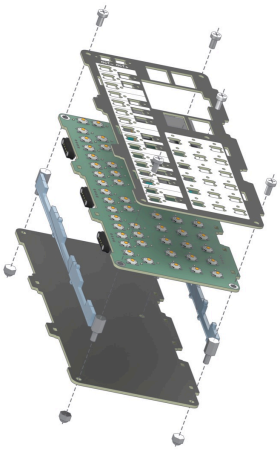
## The controls

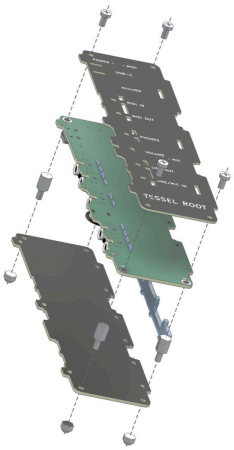
Every unit has keys down its left-hand side, a screen with two encoders above it and two below, and eighteen buttons. The screen is split into four boxes, and each encoder changes the box beside it. This booklet numbers them across and down: encoder 1 top left, 2 top right, 3 bottom left, 4 bottom right. They turn, push and hold. Anything that wipes or replaces asks you to hold, while a bar fills.

**Look after your ears.** Turn the volume wheel down before you put headphones on, then bring it up.

# Build them

The Root and the units go together the same way; a unit also has its screen.






1. Lay the top panel on the board, printing up.
2. Push a screw down through each corner hole and the middle one. Holding them in, turn it over and screw a post onto each, finger-tight.
3. **Units only:** turn it back. Cut two short strips from the double-sided tape and stick them carefully on the back of the screen. Peel off their backing, feed the screen's ribbon cable down through the slot beside the square window, and press the screen into the window. Turn it over, lift the dark latch on the connector by the slot, push the cable fully in, and press the latch down.
4. Stand the walls in their slots along the long edges: two on a unit, one on the Root's right. Each fits only its own side.
5. Lay the base on, its holes over the posts and its slots over the walls. Put a domed nut on each corner post and the plain nut on the middle one. Hold each screw with the screwdriver and tighten its nut with the spanner: firm, not forced.

No soldering and nothing to glue. If a part won't go, it's the wrong way round. The magnets are already inside the gold contacts on the sides.

# Join them up

1. Put the Root on the left. Slide a unit against its right-hand side: the magnets pull it home. Add more the same way, up to eight units, in any order.
2. Plug the power supply into the Root's **USB-C**. The POWER light comes on, and so does each unit the supply can run. A unit that stays dark needs a stronger supply.
3. Plug headphones into **PHONES**.

Units come apart and go back together whenever you like, even while you play, and each remembers everything: its sound, its page, its settings.

## The Root's sockets

Down its left-hand side, from the top:

| | |
|---|---|
| **USB-C** | Power in. Plugged into a computer, it's also a MIDI device (the computer plays your units) and the way updates come in. |
| **microSD** | A memory card, for more samples and for backups. |
| **MIDI IN** | From a keyboard with a 3.5 mm MIDI out, or a five-pin one through a Type A adapter. |
| **MIDI OUT** | To other instruments or a computer. |
| **PHONES** | Headphones, with the volume wheel just below. |
| **LINE OUT** | To speakers, a mixer or an audio interface, always at full level. |
| **LINE/MIC IN** | Sound in, for sampling and effects. There's a small microphone built in too. |

The Root has no buttons: you set it from any unit's SETUP, the bottom-right button, pressed again for the Root's part. Each unit gets its own MIDI channel automatically.

## Choose what a unit becomes

A new unit is blank, and can become any of the ten (page 7). Join it to the Root and switch on.




1. Turn encoder **1** to choose.
2. Hold encoder **4** until the bar fills. The Root gives the unit its firmware, and it starts up as its new self.
3. Write its button names on the white panel. Every unit's are in the manuals and on the website.

**Testing it:** before step 1, if you like, push encoder **3** (TEST) and follow the screen. It checks every button, light, colour, encoder and join, and the sound. A guide: **[web address to come]/test**.

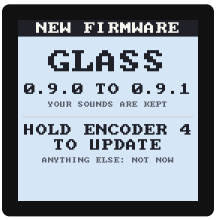
**To change it later:** press SETUP until the RESET page, turn encoder **1** to REFLASH and encoder **3** to the new kind, and hold encoder **4**. Sounds you made stay on the Root, ready if you change back.

## Update the firmware

The Root holds the firmware for every kind of unit. To get the newest:

1. On a computer, download the latest firmware: one file, for everything (page 7).
2. Take every unit off the Root, then plug its USB-C into the computer. On its own, the Root appears as a drive called **TESSEL**.
3. Drag the file onto the drive. The MIDI light flickers while the Root checks and stores it; when the drive comes back, it's done (if the computer says it wasn't ejected properly, that's fine). If the cable comes out halfway, the Root carries on as it was: start again.

Unplug the Root and snap the units back on. Each one asks. Hold encoder **4** to update it, or press anything else for later. Your sounds are kept.

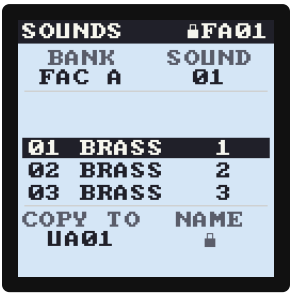



**Not settled yet:** the website is still being built.

# Play

The keys run down the left-hand side, lowest note at the top. To play them like a piano, turn the row a quarter-turn anticlockwise: low notes on the left. OCT- and OCT+ shift them an octave.

## Find the sounds




1. Press SOUNDS, the bottom middle button.
2. Turn encoder **1** for a bank and encoder **2** for a sound, and play.

Strike's kits have buttons of their own: BOOM, PUNCH and the rest.

The other buttons open pages of settings, four at a time on the encoders. Turn anything and you'll hear it. A factory sound is never overwritten: a star in the corner means you've changed it, and choosing it again puts it back. Keeping sounds of your own is in *Making Sounds*.

**Haze** puts effects on the rest: SOUNDS for a preset, then SENDS for how much of each unit goes in.

**Step** plays all the others, one track per part. *Making Music* starts with it.

## The range

| | |
|---|---|
| **Glass** | An FM synth: bells, electric pianos, glassy pads, hard basses. |
| **Volt** | A mono synth for bass and lead, with an acid mode. |
| **Ember** | A six-voice synth for chords and pads, with chorus. |
| **Axis** | A vector synth: four waves, blended and moved over time. |
| **Strike** | A drum machine: six synthesised kits, and kits of your own. |
| **Taste** | A sampler: record, chop and play. |
| **Grit** | A granular synth: any sound into clouds and textures. |
| **Haze** | Effects for the whole row: reverb, echo, chorus, drive. |
| **Step** | A sequencer: sixteen tracks, playing every other unit. |
| **Pilot** | A controller: knobs from any units on one page, and scenes. |

## Where next

Two free books take it from here, step by step, with sounds to hear:

- *Making Sounds*, if you have one unit: how synthesis works, one kind of unit at a time.
- *Making Music*, if you have several: techno, acid, house, hip-hop, ambient and more, written by you on Step, Strike and Volt to begin with.

QR code to come

The books, the firmware and every unit's button names: **[web address to come]**

No account or subscription, ever.

## If something's not right

| | |
|---|---|
| **No sound** | Check the volume wheel, and that the headphones are in PHONES. |
| **A unit is dark** | The supply can't run it, or it's a ninth unit. |
| **Something odd** | Run the test: SETUP, the RESET page, TEST. It checks every button, light and colour, and the joins between units; your sounds are kept. The guide at **[web address to come]/test** walks you through it and what each result means. |
| **Start afresh** | SETUP, the RESET page, SETTINGS (keeps your sounds) or EVERYTHING; hold encoder **4**. |

## Safety

- Not suitable for children under 3: small parts.
- Keep it dry and away from heat.
- Use a USB power supply in good condition, made for the purpose.
- Loud sound damages hearing: keep the volume moderate, especially on headphones.

[Conformity marks (UKCA, CE, FCC), the recycling symbol and the regulatory wording go here once tested.]

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Contact details to come. Draft, 1 November 2026.